

Ultimate gift guide for the CERN community

Looking for gift ideas? Here are 13 on-site options to find the perfect present



Looking for gift ideas? Look no further. Here are 13 locations at CERN where the CERN community can find the perfect present.

1. CERN shop

Attracted by the magnetic pencils? Loving the Lagrangian notebooks? The CERN shop is the place for you. Open Tuesday to Sunday from 8 a.m. to 6 p.m. at the CERN Science Gateway reception, the shop will remain open for most of the end-of-year closure. Note that CERN personnel receive a 10% discount during December when showing their CERN access card.

2. CERN Library bookshop

Get tempted by literary treats in the new-look CERN Library bookshop (52/1-054), open Monday to Friday from 9 a.m. to 6 p.m.

3. CAGI cultural kiosk

Get [discounts](#) for sports, culture and leisure activities, including 30% off cinema tickets, 20% off Vitam UCPA, 15% off Grand Theatre of Geneva, 10% off Choco Passes, plus a special offer for Glacier 3000. The CAGI kiosk in CERN's Main Building is open from 9 a.m. to 1.30 p.m. and from 2 p.m. to 5 p.m. You can also order tickets via the [CAGI website](#) using the online form at the end of each activity's page. Sign up [here](#) to receive the weekly newsletter.

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A Word from Mike Lamont

Looking back on an eventful year for accelerators

Accelerator highlights, challenges and successes in 2023

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Impartiality or, better, multi-partiality?

A Word from Mike Lamont

Looking back on an eventful year for accelerators

Accelerator highlights, challenges and successes in 2023

During their recent visit to CERN, Presidents Berset and Macron were given an introduction to the Laboratory by Fabiola. President Macron appeared to be really interested in the origins of the Universe, and explanations went way over time! The entourage were then whisked down to the LHC tunnel and through to the ATLAS cavern. It's hard not to be impressed, the LHC is the LHC and ATLAS is awe inspiring. What they didn't see, however, was the near miraculous demonstration of technological prowess that underpins all of this. Taking a step back, it's quite remarkable what we do here. One can weave a number of threads through the accelerator complex, from the chopper of Linac 4 to the PS RF system, the multi-cycling synchronisation of everything, the production of radioactive ion beams, the vacuum systems from ELENA to the LHC, the n_TOF target, stochastic cooling of antiprotons, the power converters, the magnets (from 1959 to 2023), beam instrumentation, the LHC transverse feedback system... and but wonder that it all comes together as well as it does.

It's worth reflecting on this, as we look back on another year in the life of an unparalleled collection of accelerators and facilities, a year that has been very good overall, although somewhat eventful for the LHC.

Despite the sophisticated operations involved in producing the multiple beam configurations required for the down-stream machines, Linac4 maintained an impressive 98% availability, with stable running and optimal beam performance. The Proton Synchrotron Booster contributed significantly, as always, supplying beams to ISOLDE, HIE-ISOLDE and MEDICIS, as well as to the PS for the downstream users – all within tight user-dependent specifications. A large fraction of the protons at CERN are sent to ISOLDE, with 11.3×10^{19} protons heading that way in 2023 and around 4.5×10^{19} going on to the PS. To quote Erwin Siesling: "We (ISOLDE) had yet another very successful year, full of the usual issues and problems but with great physics results and lots of happy users!"

The PS delivered beams to n_TOF, AD-ELENA, and the East Area experiments and irradiation facilities, which include CLOUD, CHARM, and IRRAD. The AD, back online since 30 July after repairs to a faulty quadrupole, compensated for the late start by extending their run to 12 November. Following optimisation, the AD achieved record intensity antiprotons for ELENA and the experiments.

Throughout 2023, the SPS operated very well, with no major faults or prolonged downtimes while achieving an impressive transmission rate of about 95% to the North Area experiments with well optimised beam quality.

Besides delivering beam to their regular users, work has continued in the Injectors on the high-intensity, high-brightness beams required by the HL-LHC (the primary mission of the LHC Injector Upgrade (LIU) during LS2). The injector teams have done a great job, with the LHC beams from PSB, PS and SPS meeting the LIU target beam parameters and even showing some margin to surpass the beam intensity and brightness required by HL-LHC. In the first part of the year, the LHC demonstrated outstanding luminosity performance, both peak and integrated. Operationally the teams have established impressive flexibility and sophisticated operational and system-level control. However, the excellent availability was punctuated by some singular faults – in particular, a helium leak into the insulation vacuum of the inner triplet assembly left of point 8 in the middle of July. This was a serious event, but reactivity was fantastic, and the leak repair and all that went with it were widely seen as a remarkable collaborative recovery.

The adaptability of the cryogenics team was key to avoiding the need to warm up the adjacent sector. The leak, in an edge-welded bellows, was the result of a quench caused by an electrical disturbance on the grid. An availability analysis will be conducted at the Chamonix meeting in 2024 to address other potential non-conformities dating from construction. The prompt recovery enabled some special runs and the first LHC ion run in five years.

Lead ions at the end of the year are always interesting, with preparation of the ion source, Linac3, and LEIR starting months before beam is sent to the PS and the downstream machines.

Ions are principally destined for the North Area and the LHC. However, in the last two weeks of the four-week run, the PS provided lead ions to the East Area, where the CHIMERA facility irradiates electronics with high-energy heavy ions to study the effects of cosmic radiation on the electronics used in the CERN accelerators and experiments, as well as for space missions and avionics.

In the SPS, the first operational use was made of a technique known as “momentum slip-stacking”, which involves injecting two batches of four lead-ion bunches separated by 100 nanoseconds to produce a single batch of eight lead-ion bunches separated by 50 nanoseconds, an impressive example of “RF gymnastics” and low-level RF control.

In the LHC, the lead nuclei were colliding this year with an increased energy of 5.36 TeV per nucleon pair (compared to 5.02 TeV previously). A record number of bunches and high bunch intensities – thanks to the downstream machines – made for a challenging ion run. Again, with concerted effort and adaptability, the teams wrestled down the issues, delivered some record performance and paved the way for the rest of Run 3.

As we look back on the year, we should bear in mind the phenomenal job that’s done in the exploitation of the complex. This is difficult stuff and it’s remarkable that it all works as well as it does. President Macron might not have seen it, but he surely sensed the spirit.

Mike Lamont

Director for Accelerators and Technology

Ultimate gift guide for the CERN community

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We're pulling our socks up and have researched 13 different options for gift ideas for the CERN community. (Image: CERN)

4. Kiosk

Located in CERN's Main Building, the kiosk stocks not only CERN-branded chocolates, but also CERN-branded Swiss-army knives. Open Monday to

Friday from 7.30 p.m. to 11 a.m. and 11.30 a.m. to 4 p.m.

5. Restaurants

The CERN restaurants offer a range of edible gift ideas. Restaurant 1 in CERN's Main Building will also host its own Christmas market on the 12, 13 and 14 December 2023.

6. ATLAS experiment shop

Knock their socks off with ATLAS socks, rainbow lanyards and more. Check out the items at the [ATLAS shop](#) located at the ATLAS secretariat (40/4-D01), open Monday to Friday from 8.30 a.m. to 12.30 p.m. and 1.30 p.m. to 5.30 p.m.

7. LHCb experiment shop

Books, mugs and more, the [LHCb shop](#) is located at the LHCb secretariat (2/1-024), open Monday to Friday from 8.30 a.m. to 12.30 p.m. and 1.30 p.m. to 5.30 p.m.

8. CMS experiment shop

New hoodies and detector umbrellas are just some of the items at the [CMS shop](#), also located in Building 40, at the CMS secretariat (40/5-B01), open Monday to Friday from 9 a.m. to 12 noon and 2 p.m. to 5 p.m.

9. CERN alumni offers

CERN personnel past and present can join the [CERN alumni](#) network and take advantage of [partner offers](#) including hotel discounts.

10. Staff Association offers

From discounted ski passes and sports equipment to Aquaparc offers, members of the Staff Association can enjoy the full list of offers available through the [Staff Association partnerships](#).

11. CERN Marketplace

Give a pre-loved item a new home – browse what's on offer at the [CERN Marketplace](#).

12. CERN & Society Foundation

Make a difference by donating to one of the many projects of the [CERN & Society Foundation](#).

13. Staff Association fundraising

Give to those in need by [supporting](#) the Staff Association's long-term collections.

CERN Community: take part in the CERN end-of-year crossword!

Submit your answers by 8 January and you could win an LHC Lego set, two Geneva Choco passes or two Balexert cinema tickets

CERN has had a busy 2023. If you were paying attention, or if you fancy testing your knowledge, why not take part in the CERN 2023 end-of-year crossword? You could be in with a chance of winning a full LHC **Lego set** (worth 98 CHF), **two CAGI Choco passes** (worth 28 CHF each) or **two CAGI Pathé Balexert cinema tickets** (worth 14 CHF each). *Please note: You must have a CERN email address to enter the prize draw.*

Interested? Here's how to enter:

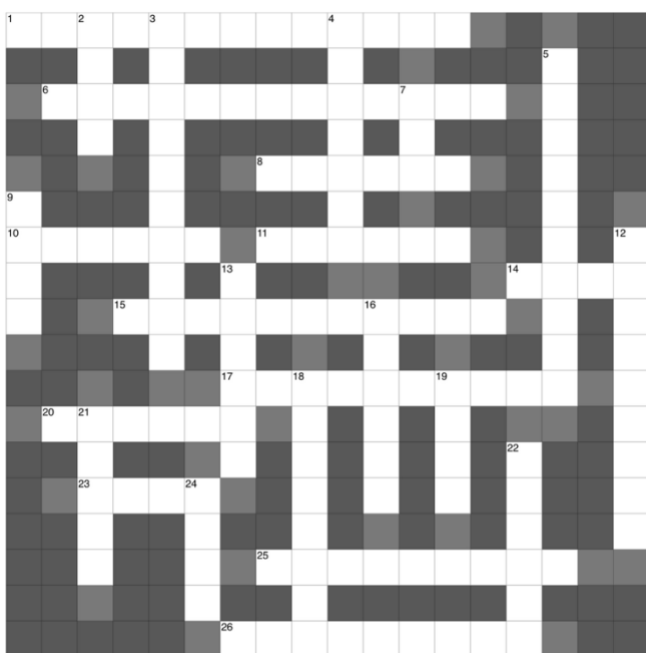
- See below or [download this PDF](#) to access the crossword

- Send your answers **from your CERN email address** to bulletin-editors@cern.ch **before 8 January 2024**. Your answers can be an attached completed crossword or a numbered list.
- **All correct entries** will be entered into a prize draw. **The prize winners will be announced in the first Bulletin of 2024.**

Please note: you must have a CERN email address to enter the prize draw.

Good luck!

CERN Internal Communication



ACROSS

1. What was the subject of the third edition of the Sparks! Forum held at CERN? (6, 7)
6. Which new tool allows CERN's on-site personnel to look at news, a map, the phonebook and the restaurant menus? (4, 6, 3)
8. In September, the ATLAS experiment released a paper that had a high precision measurement of which of the four fundamental forces? (6)
10. Launched this year, which space telescope relies on CERN's computing infrastructure to investigate the effects of dark matter and dark energy on the Universe? (6)
11. CERN's very own astronaut started his training at ESA this year. What country is he from? (6)
14. CERN and which other well-known scientific institution got together this July to discuss open science? (4)

15. The newly-inaugurated CERN library is home to the Scientific ____ Service? (11)

17. Indico surpassed this number of events on its platform this year (3, 7)

20. In May, the ATLAS and CMS experiments got together to present evidence of a rare Higgs boson decay into a Z boson and which other particle? (6)

23. Towards the end of 2023, the ALICE experiment benefited from the first heavy-ion run in five years. Ions of which element circulated in the LHC? (4)

25. In April, CERN's Knowledge Transfer group established a partnership with an Irish company called Supernode to collaborate on finding solutions to the transfer of which type of energy? (9)

26. Which film festival takes place on the CERN campus? (9)

DOWN

2. This object, about 55 km from CERN, caused an electrical disturbance that brought the LHC to a halt during the summer (4)

3. Science Gateway, CERN's brand new centre for education and outreach, was designed by this architect (5, 5)

4. 2023 marked 50 years since the discovery of these types of currents (7)

5. Arts at CERN awarded Joan Heemskerk the Collide residency, which, this year, was split between CERN and which European capital city? (10)

7. The Feasibility Study for the Future Circular Collider started gathering data on the ground in the Canton of Geneva, the French department of Haute-Savoie and which other department of France? (3)

9. Every year, the CERN accelerator complex undergoes a pause for upgrades before restarting in the new year. Which acronym is this pause known by? (4)

12. In line with Einstein's predictions, the ALPHA experiment found that antimatter does what under the influence of gravity? (5, 4)

13. W and Z, discovered at CERN exactly 40 years ago, are what types of particles? (6)

16. 2023 marks how many years since CERN released the World Wide Web to the public? (6)

18. To which future telescope, designed to detect gravitational waves, is CERN contributing its expertise in vacuum technology? (8)

19. In June, which LHC collaboration released its latest results on the asymmetry between matter and antimatter? (4)

21. The civil engineering for the next upgrade of the LHC has now been fully completed. How is this upgrade often referred to? (5)

22. Which CERN facility is investigating the creation of a nuclear clock? (6)

24. The CMS experiment recently presented results on a search for exotic particles. This particular study focused on the possible production of long-lived particles, including which kind of exotic photons? (4)

CERN publishes its environment report for 2021–2022

The report highlights progress and areas for future development in CERN's ongoing efforts to minimise its environmental footprint

Today, 4 December 2023, CERN is publishing its third environment report for the years 2021 and 2022.

This report chronicles the advancements made in key environmental domains and specifically covers the measures taken by CERN to reach its priority objectives: limiting the rise in electricity and water consumption and reducing direct greenhouse gas emissions.

During a large part of this reporting period, CERN's accelerator complex was undergoing a long

shutdown for maintenance and upgrades. In July 2022, operations successfully resumed with the start of Run 3. CERN is well on track to meet the reduction objectives set for the end of Run 3 at the end of 2025, which use 2018 as the baseline year. "Over the years since we embarked on our first environment report, we have learned a great deal about our environmental footprint, implemented mechanisms to better understand and control it and increased our efforts to identify and develop technologies stemming from our core research

that have the potential to benefit the environment,” says Benoît Delille, head of the CERN Occupational Health and Safety and Environmental Protection unit.



CERN Science Gateway is a net-zero carbon footprint building. (Image: CERN)

The Laboratory is committed to limiting the rise in electricity consumption to 5%, corresponding to a target of 1314 GWh, while significantly improving the performance of its facilities. A total of 1215 GWh was consumed in 2022 and the accelerator complex is now more efficient. In the light of the energy crisis, CERN also implemented additional energy-saving measures as a mark of social responsibility, and further explored the diversification of energy sources and the launch of heat-recovery projects. Thanks to these efforts, CERN was awarded the internationally recognised ISO 50001 energy-management certification earlier this year.

CERN is committed to keeping the increase in its water consumption below 5% despite a growing demand for water cooling at its facilities. Since 2000, CERN has radically decreased its water consumption by about 80%.

With respect to direct greenhouse gas emissions, CERN's objective is to achieve a 28% reduction by the end of Run 3, which corresponds to a target of 138 300 tCO₂e. In 2022, direct emissions of 184 300 tCO₂e were generated. A comprehensive programme to ensure progress towards the objective is in place, which includes increased efforts by the experiments to limit gas

consumption and work towards replacing current gases with more environmentally friendly ones.

When reporting on indirect greenhouse gas emissions, CERN now includes emissions arising from procurement and details the main sources of related emissions.

The report also explores other key environmental areas, including how waste is managed. In 2022, the recycling rate for conventional waste was 69%, compared to 56% in 2018.

Biodiversity, land usage and landscape change are another important focus of the report. Among several activities carried out during 2021 and 2022, CERN took flora and fauna inventories and identified “areas of interest” to be classed as a priority for preservation in the event of any planned works or projects.

Over the reporting period, noise levels measured at the boundaries of CERN's sites were typically around 50 dBA during the day and 45 dBA at night. CERN has implemented an online real-time monitoring system at specific points on its site.

With respect to ionising radiation, the European annual dose limit for public exposure to artificial sources is 1 mSv. CERN is committed to keeping its contribution to no more than 0.3 mSv per year. In 2022, the actual dose received by any member of the public living near the Laboratory was less than 0.01 mSv, which is more than 100 times lower than the average annual dose received from medical exposure per person in Switzerland.

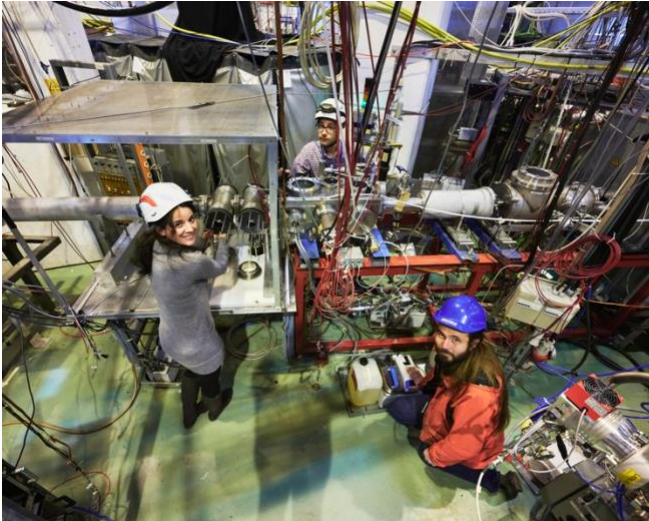
Finally, the report covers the latest information on how CERN's technology and knowledge benefit society, notably via the new CERN Innovation Programme on Environmental Applications launched in March 2022.

CERN is committed to environmentally responsible research and transparent reporting. As a leading scientific organisation, CERN aspires to be a role model, publishing biennial environment reports that adhere to the Global Reporting Initiative (GRI) reporting standards.

The online version of the report is available at: <https://hse.cern/environment-report-2021-2022>. A PDF version is available [here](#).

Exotic atomic nucleus sheds light on the world of quarks

CERN and the Accelerator Laboratory in Jyväskylä, Finland, have teamed up to weigh in on the understanding of the effects of the weak force on quarks



The ISOLDE set-up used to study the exotic nucleus of aluminium. (Image: CERN)

Experiments at CERN and the Accelerator Laboratory in Jyväskylä, Finland, have revealed that the radius of an exotic nucleus of aluminium, ^{26m}Al , is much larger than previously thought. The result, described in a [paper](#) just published in *Physical Review Letters*, sheds light on the effects of the weak force on quarks – the elementary particles that make up protons, neutrons and other composite particles.

Among the four known fundamental forces of nature – the electromagnetic force, the strong force, the weak force and gravity – the weak force can, with a certain probability, change the “flavour” of a quark. The Standard Model of particle physics, which describes all particles and their interactions with one another, does not predict the value of this probability, but, for a given quark flavour, does predict the sum of all possible probabilities to be exactly 1. Therefore, the probability sum offers a way to test the Standard Model and search for new physics: if the probability sum is found to be different from 1, it would imply new physics beyond the Standard Model.

Interestingly, the probability sum involving the up quark is presently in apparent tension with the expected unity, although the strength of the tension depends on the underlying theoretical calculations. This sum includes the respective

probabilities of the down quark, the strange quark and the bottom quark transforming into the up quark.

The first of these probabilities manifests itself in the beta decay of an atomic nucleus, in which a neutron (made of one up quark and two down quarks) changes into a proton (composed of two up quarks and one down quark) or vice versa. However, due to the complex structure of the atomic nuclei that undergo beta decays, an exact determination of this probability is generally not feasible. Researchers thus turn to a subset of beta decays that are less sensitive to the effects of nuclear structure to determine the probability. Among the several quantities that are needed to characterise such “superaligned” beta decays is the (charge) radius of the decaying nucleus.

This is where the new result for the radius of the ^{26m}Al nucleus, which undergoes a superallowed beta decay, comes in. The result was obtained by measuring the response of the ^{26m}Al nucleus to laser light in experiments conducted at CERN’s ISOLDE facility and the Accelerator Laboratory’s IGISOL facility. The new radius, a weighted average of the ISOLDE and IGISOL datasets, is much larger than predicted, and the upshot is a weakening of the current apparent tension in the probability sum involving the up quark.

“Charge radii of other nuclei that undergo superallowed beta decays have been measured previously at ISOLDE and other facilities, and efforts are under way to determine the radius of ^{54}Co at IGISOL,” explains ISOLDE physicist and lead author of the paper, Peter Plattner. “But ^{26m}Al is a rather unique case as, although it is the most precisely studied of such nuclei, its radius has remained unknown until now, and, as it turns out, it is much larger than assumed in the calculation of the probability of the down quark transforming into the up quark.”

“Searches for new physics beyond the Standard Model, including those based on the probabilities of quarks changing flavour, are often a high-precision game,” says CERN theorist Andreas Juttner. “This result underlines the importance of

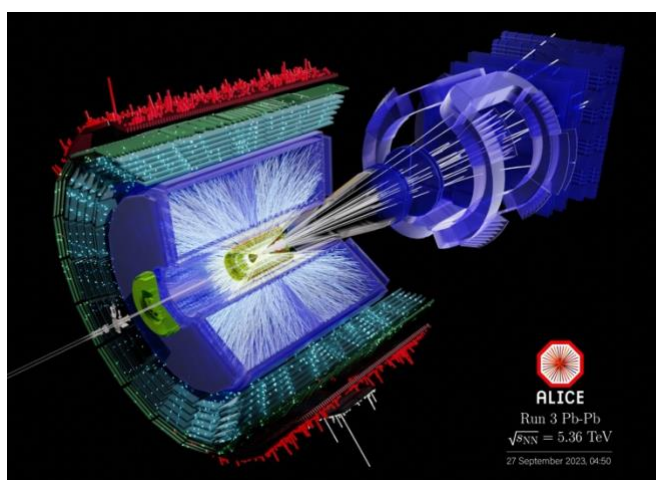
scrutinising all relevant experimental and theoretical results in every possible way.”

Past and present particle physics experiments worldwide, including the LHCb experiment at the Large Hadron Collider, have contributed, and are continuing to contribute, significantly to our knowledge of the effects of the weak force on

quarks through the determination of various probabilities of a quark flavour change. However, nuclear physics experiments on superallowed beta decays currently offer the best way to determine the probability of the down quark transforming into the up quark, and this may well remain the case for the foreseeable future.

ALICE bags about twelve billion heavy-ion collisions

The whopping number of collisions recorded by ALICE during the recent five-week heavy-ion run of the LHC is 40 times greater than the total recorded by the experiment in its previous periods of heavy-ion data taking, from 2010 to 2018



A lead–lead collision event in the ALICE detector. (Image: CERN)

After a five-year pause, on the evening of 26 September, lead ions collided at the Large Hadron Collider (LHC) at an unprecedented high energy of 5.36 TeV per pair of nucleons (protons or neutrons) and a collision rate six times higher than before. The final lead-ion beam of this latest heavy-ion run was dumped early in the morning of 30 October, after a forced magnet ‘quench’, carried out to better understand the amount of deposited energy at which the LHC superconducting magnets lose their superconducting state. This improved understanding of the LHC machine will help to further increase the heavy-ion collision rate in the near future.

For this much-anticipated heavy-ion run, alongside improved beam parameters, the ALICE experiment – the LHC’s heavy-ion specialist – made use of its significantly upgraded detector

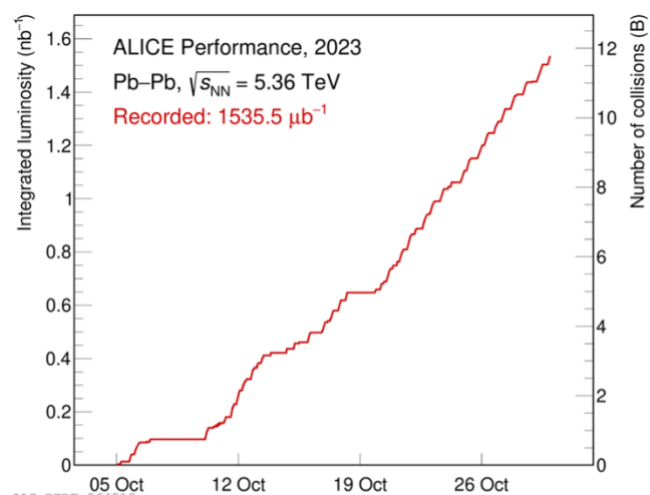
with continuous readout electronics. This means that each and every collision can now be recorded and is thus available for physics analysis, whereas, in the past, only a fraction of collisions could be selected for recording. This continuous readout was achieved by revamping the experiment’s time projection chamber (TPC) detector and upgrading the readout electronics of all of the detectors. In addition, the new inner tracking system (ITS) detector, which is based on highly granular silicon pixel technology, provides sharp images of the collisions with its 10 m² of active silicon area and nearly 13 billion pixels within the three-dimensional detector volume.

The resulting dramatic increase in the data rate was facilitated by the deployment of a new computing infrastructure for online data processing. This infrastructure includes a new data processing farm that sends the data produced by the experiment directly to CERN’s Data Centre, located about five kilometres from ALICE, through a dedicated high-speed optical-fibre connection that had to be established to cope with the increased data rate.

During the five-week run, ALICE recorded about 12 billion lead–lead collisions – 40 times more collisions than the total recorded by ALICE in the previous periods of heavy-ion data taking, from 2010 to 2018. The new data processing farm, consisting of 2800 graphics processing units (GPUs) and 50 000 central processing unit (CPU) cores, routinely digested collision data at a rate of up to 770 gigabytes per second. It then compressed the data to about 170 gigabytes per

second before shipping it to the Data Centre for storage on disk and later, at a limited speed of 20 gigabytes per second, for storage on tape for long-term preservation.

The fresh data set – which amounts to 47.7 petabytes of disk space and is now being analysed – will advance physicists’ understanding of quark–gluon plasma, a state of matter in which quarks and gluons roam around freely for a very short time before forming the composite particles called hadrons that ALICE detects. The increased number of recorded collisions will allow the ALICE researchers to determine the temperature of the plasma using precise measurements of thermal radiation in the form of photons and pairs of electrons and positrons. It will also allow other properties of the nearly-perfect fluid to be measured with greater precision, especially using hadrons containing heavy charm and beauty quarks.



The number of lead–lead collisions collected by ALICE in 2023, expressed in terms of the cumulative number of collisions (right vertical axis) and a related quantity called integrated luminosity (left vertical axis). (Image: CERN)

ALICE collaboration

HiLumi News: Successful test paves the way for magnet production at CERN

Engineers have successfully tested an improved design of a quadrupole magnet manufactured at CERN, confirming that niobium–tin technology is viable for accelerator magnets



The magnet, named MQXFB03, is 7.2 metres in length. It is the first of ten that will be needed for the HL-LHC. (Image: CERN)

The Large Hadron Collider (LHC) needs specific types of magnets to tightly control the beams of particles at its collision points. Called final-focusing quadrupoles, these magnets are installed in the LHC’s interaction regions around the experiments. For the high-luminosity upgrade of the LHC (HL-

LHC), the final-focusing magnets at ATLAS and CMS will need replacing. Tests at CERN have now confirmed that the quadrupole magnets newly designed to replace them will work.

Unlike the LHC magnets, which are made from niobium–titanium (Nb–Ti), the new magnets are made from a more challenging material: niobium–tin (Nb₃Sn). “Given the brittleness of Nb₃Sn and the fact that its coils are very rigid, assembling Nb₃Sn magnets requires close attention,” explains José Miguel Jiménez, head of the Technology department. “This makes it a much bigger challenge than for Nb–Ti magnets.”

The CERN Technology department is developing a series of ten magnets (eight, plus two spare), each 7.2 metres in length. This work builds on the HL-LHC Accelerator Upgrade Project (AUP), based in the USA, which is currently manufacturing 20 (16, plus four spares) quadrupole magnets, each 4.2 metres long. Recent tests at Fermilab showed

that these magnets operate at target current at both 1.9 kelvin (-271.25 °C) and 4.5 kelvin (-268.65 °C), thus meeting the project requirements. The CERN team is relying on the same design and similar manufacturing procedures as AUP but scaling them up to 7.2-m-long magnets.

“The contribution of our US colleagues has been instrumental in developing the design and procedures for these magnets, and the regular cross-checks of manufacturing and test data have helped the teams on both sides of the Atlantic to overcome many challenges,” says Ezio Todesco, who is in charge of the HL-LHC interaction region magnets.

The successful test at CERN, which ran from August to October, achieved the target current of 16.53 kA at both 1.9 K and 4.5 K. The target current corresponds to the 7 TeV LHC operation, plus a 300 A margin. Although operation is planned at 1.9 K, the ability to reach target current at 4.5 K confirms design robustness and a comfortable operation margin for the HL-LHC and beyond.

This is the third full-length magnet to be tested as part of a recovery plan decided on after

performance limitations were observed on the first two prototypes. The other magnets showed no signs of degradation when tested, but were always limited to below target current when operated at 4.5 K. The team at CERN paused production to investigate this limitation. By improving the design of the outer shell, reducing peak stress on the magnet during coil assembly and changing the parameters of the coil manufacturing process, they eliminated the limitations and the third magnet has outshone its predecessors.

“Thank you to all the contributors for the excellent results and efficient teamwork and for deriving practical and robust engineering solutions to bring niobium–tin technology to the maturity level required for accelerator magnet applications,” says Arnaud Devred, the TE-MS group leader.

“This is a fantastic result for the project,” says Oliver Brüning, HL-LHC project leader. “It means that niobium–tin is viable for 7-m-long accelerator magnets and is an enabling technology for HL-LHC.”

TE department

Ethical, open and non-commercial: the Open Web Search project is designed to provide Europe with the right alternative to existing search engines

CERN recently hosted the fifth International Open Search symposium, where over 100 experts discussed how to make searching the web in Europe fairer and safer

In October 2023, CERN hosted the fifth International Open Search symposium. Attending the symposium were more than 100 experts, comprising representatives of 60 institutes and universities, as well as representatives of industry and policy makers, who got together to discuss how to make searching the web fair, safe and less biased across Europe. This symposium was the latest event in the framework of a project that has grown organically over many years, with its roots in a workshop held in Munich in 2017. Called the Open Web Search project, it was launched by a consortium of 14 research partners with the goal of securing Europe’s contributing to Europe’s

digital sovereignty as well as promoting an open human-centered search engine market.

“Globally, web search is in the hands of five players,” explains Andreas Wagner, a CERN IT department member involved in the project. “The role of CERN within the Open Web Search project is to lead the design and implementation of the required distributed data infrastructure.”

Indexing the huge amount of information currently available on the web is a massive endeavour, but doing it in an unbiased way is an even harder task. “Together with the other partners, we have started by simply discussing possible ways of building a new neutral indexing

system,” explains Andreas. “Although the system is still very preliminary, running the it on our own set of webpages at CERN has proved useful as it has allowed us to learn critical things about our own internal search engine. In other words, the project will also help CERN to improve its own search capabilities and will provide an open science search function across CERN's multiple information repositories.”



(Image: openwebsearch.eu / using images by NASA, Unsplash (christopher-burns-dzejyfCAzIA-unsplash), OpenClipart-Vectors from Pixabay (1293096))

To build a search engine, you first need to create a bot that will download the entire content of web pages. An algorithm is then run on the downloaded pages to build the actual index. If the index is biased, it can add visibility to certain pages and reduce the visibility of others.

“Indexing basically means taking a snapshot of the digital world, and so copyrights and legal and ethical matters enter into play,” says Andreas. Ethics is a key element of the whole Open Web Search project. The currently available search engines provide the user with a list of results that are presented according to the tailored algorithm behind the whole process. “Everything is designed to make the user happy, so only a selection of relevant documents are shown,” confirms Andreas. “Commercial search engines use profiling techniques to present tailored results, and this raises ethical and privacy concerns.”

To keep the new algorithms neutral and transparent, the team is considering solutions that do not profile the individual user but rather look at the behaviour of anonymised groups of users to return meaningful results.

“Current commercial search engines deliberately make you stop here and there and divert your attention because this serves their commercial interests,” continues Andreas. “We want to take any user straight to their destination in a fair, effective and transparent way.”

The project OpenWebSearch.EU has received funding from the European Union's Horizon research and innovation programme under grant agreement No 101070014.

Antonella Del Rosso

Celebrating 25 years of service to CERN in 2023

Staff members who marked 25 years of service to CERN in 2023 were invited by the Director-General on 2 November to the traditional ceremony held in their honour.

The photos from the ceremony and the list of the 90 staff members concerned can be viewed in this [album](#) (restricted access).

We thank them all warmly for their commitment and wish them continued success at CERN!

HR department

Computer Security: Four gifts for your Christmas break

Just as physical threats don't stop outside working hours, CERN's annual closure won't put a stop to cybersecurity attacks in the coming weeks. The number of attacks might even increase, as attackers* are aware that people are more relaxed during this time, less alert, probably on holiday, or blissfully ignorant of the possibility of evil during a time that's supposed to be peaceful (but unfortunately rarely is). CERN remains on the attackers' radar and we must ensure that our protective systems are up and running, improving and enhancing – as also suggested by the recent audit of CERN's cybersecurity (more on that in a future *Bulletin* article).

So, with the Christmas period fast approaching, let's have a look at what's been put under the tree by Security Claus and his protective elves (assuming you've behaved well and deserve it):

- [Xorlab "ActiveGuard" and Microsoft "MDO" mail filtering](#). The basic bring-along: chocolate, pralines and bonbons. But, unlike sugary treats, ActiveGuard and MDO are a protective diet, filtering away all malicious and spammy emails. Having two quarantine portals may seem unnecessary, but two is better than none. It's like chocolate: you can never have enough.
- [Threatray and ESET anti-malware](#). An awesome gift card for you and your family: download CERN's new anti-malware software for free and install and run it on your personal laptop, Macbook and at home. We've already deployed the

sophisticated Threatray inspector on all CERN-managed Windows devices, and ESET will come along next year to protect your work even better.

- [Multifactor protection](#). Like ties and socks, this may not be your favourite present, but it's very effective: the silver bullet you need to protect your account against password theft/loss/disclosure.
- [Gitlab security scanners](#). For those who like puzzles (and do programming): the Gitlab security scanners (static application security testing, or SAST, as well as a module for finding clear text passwords and other credentials) provide you with essential means to identify misconfigurations, vulnerabilities and bugs. We run them for you – all you need to do is check their output. And put the puzzle pieces in place.

With those four lovely and protective presents, we wish you a calm and relaxing annual closure, joyous festivities and a happy new year. Stay healthy and safe. And thanks in advance if we can count on you to help keep CERN secure in 2024 as well! Cheers!

** Not to mention the friendly hackers from the "German Chaos Computer Club" who hold their "Chaos Communication Congress" at the end of each year and, unsurprisingly, regularly knock on CERN's firewall, probing us for vulnerabilities ... Ho, ho, ho!*

The Computer Security team

Official news

Service availability during CERN's annual closure 2023/2024

General site services

As always, in addition to the Fire and Rescue service (+41 22 76) 74444, the Security service

remains operational every day (24/7) and can be reached at (+41 22 76) 76666.

SCE department services that do not depend on continuous human presence will remain available, although at a reduced support level. In general, the response time for common issues will be half a day (but this is not guaranteed). To report urgent infrastructure issues during the annual closure, call the CCC TI operator at (+41 22 76) 72201.

Other services requiring human presence (such as Service Desk, CERN hostels, goods reception, shuttles, cleaning services, etc.) will not operate during the end-of-year closure.

The self-service [Car Sharing](#) and [Bike Sharing](#) services will remain available.

For more information, please consult the [CERN Service Portal](#).

In addition, please note that the heating on the Meyrin and Prévessin sites will be switched to low-heat mode to maximise energy savings during this period of low occupancy. This will lead to a slight drop in temperature.

Computing services

Essential services provided by the IT department – including WLCG production services – will remain available.

Most problems will be dealt with on a best-effort basis only and the availability of specific services might be limited by the availability of other services. No interventions are scheduled – in case of a failure, there is no guarantee that services will be restored, and changes requiring on-site human intervention will not be possible.

Incidents will be listed on the [CERN Service Status Board for Computing](#).

The Computer Centre Operator service will be available and can be reached at (+41 22 76) 75011 where urgent problems can be reported. Calls will be redirected to the CCC TI operator during absences.

Suspected computer-security incidents must be reported to Computer.Security@cern.ch or (+41 22 76) 70500 as usual.

Please remember to shut down and power off any equipment in your office whose operation is not required during the annual closure.

CERN Health Insurance Scheme (CHIS) - Opening hours of UNIQA offices during end-of-year closure

Please note that the UNIQA office at CERN (Main Building) will be closed during the two-week end-of-year closure.

During that period, **UNIQA's offices in Geneva will remain open** daily from 8 a.m. to 5 p.m. and will be closed on Monday, 25 December 2023 and on Monday, 1 January 2024. During open periods you can also reach UNIQA by telephone on 022 718 63 00.

For **urgent medical assistance**, you may call UNIQA Assistance **+41 22 819 44 77, 24h/day throughout this period**. Please note that this service only provides medical advice and urgent assistance services and is not in a position to inform you on the coverage by CHIS of medical expenses.

HR department

Official holidays in 2024 and end-of-year closure 2024/2025

(In accordance with Articles R II 4.38 and R II 4.39 of the Staff Regulations)

Official holidays in 2024 (in addition to special leave during the annual closure):

- Monday, 1 January (New Year)

- Friday, 29 March
(Good Friday)
- Monday, 1 April
(Easter Monday)
- Wednesday, 1 May
(1 May)
- Thursday, 9 May
(Ascension day)
- Monday, 20 May
(Whit Monday)
- Thursday, 5 September
("Jeûne genevois")
- Tuesday, 24 December
(Christmas Eve)

- Wednesday, 25 December
(Christmas)
- Tuesday, 31 December
(New Year's Eve)

Annual closure of the site of the Organization during the end-of-year holiday period:

The Laboratory will be closed from Saturday, 21 December 2024 until Sunday, 5 January 2025 inclusive (without deduction of annual leave). The first working day of the New Year will be Monday, 6 January 2025.

*HR department
HR.leave@cern.ch*

Extension of the pre-retirement programmes

Following a recommendation by the Standing Concertation Committee at its meeting on 29 November 2023 and approval by the Director-General, please note that:

the Part-Time Work as a Pre-retirement Measure Scheme and the Progressive Retirement

Programme have been extended by one year, from 1 January 2024 until 31 December 2024.

Further information is available on the following links:

- [Progressive retirement programme](#)
- [Part-time work as a pre-retirement measure](#)

Safe charging of portable lithium batteries

Some precautions to bear in mind in order to minimise the fire risk from the use of lithium batteries

Fires caused by lithium batteries sporadically make the news, and CERN is not immune from this phenomenon. CERN has experienced several fires caused by a lithium battery in the course of being charged. The subsequent investigations revealed lessons for all of us.

Lithium-ion or lithium-polymer batteries are typically used in electric cars, e-bikes, computers and other power-operated equipment as well as smartphones or e-cigarettes. Even when they're small these batteries are miniature power plants. Here are some precautions to bear in mind in order to minimise the fire risk from the use of these batteries:

Things you should do:

- **Use according to the manufacturer's instructions**
Read and follow the manufacturer's instructions precisely!
- **Use safe batteries**
Make sure your batteries are safe: regularly check the condition (damage, deformation, leakage...) of the battery and immediately replace any damaged battery.
- **Use a suitable charger**
Use the charger supplied with the battery. It is designed to monitor the charge and avoid overcharging. If the original charger is not available, you may also use a CE

certified charger approved for the particular device.

- **Charge under supervision**
Supervise the charging of your batteries, in particular for powerful ones like e-bikes batteries, for example.
- **Remove battery from device**
Where batteries are designed to be removed for charging, always remove the batteries from the device before charging.
- **Unplug**
Unplug the charger once the battery is charged.
- **Dispose safely**
When not in use leave your batteries, in particular powerful ones, in a fire-proof cupboard or bag, if possible. Batteries are hazardous waste that must be disposed of following the appropriate procedures: [https://sce-dep.web.cern.ch/waste/what-goes-](https://sce-dep.web.cern.ch/waste/what-goes-where#Batteries)

[where#Batteries](#). As batteries are never fully discharged it is recommended to seal the electrical poles with tape and/or put them in fire-proof bags prior to disposal.

Things you should avoid:

- **Charge under high temperatures**
Don't charge a battery when the ambient temperature is above 35 °C.
- **Charge close to combustible materials or hazardous substances**
Do not charge batteries close to combustible materials or hazardous substances (chemicals, explosives...).
Avoid charging underground if possible.

Immediate actions you should take in case of overheating or fire

In the event your battery is overheating, swelling, melting, emitting smoke or a clacking sound evacuate the premises, close the door, warn your colleagues and call the CERN fire brigade at 74444 or +41 22 767 4444.

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

8 Dec | At CERN | CERN Science Gateway | Public events | EN

[***FULL | COMPLET*** Music and Physics: a space-time voyage back to our origins, with Yo-Yo Ma](#)

11 Dec | Computing | IT Amphitheatre | CERN Computing Seminar | EN

[PyScript and the magic of Python in the browser](#)

11–23 Dec | Knowledge Sharing | Main Building | CERN & Society Foundation | EN/FR

[Discover the CERN & Society Foundation exhibition](#)

12 Dec | Physics | Main Auditorium | LHC Seminar | EN

[Probing the \(anti-\)hypertriton properties with ALICE at the LHC](#)

14 Dec | Knowledge Sharing | CERN Science Gateway | CERN Open Science & UNESCO | EN
[Opening science for all](#)

15 Dec | Knowledge Sharing | CERN Science Gateway | IYBSSD | EN
[Closing Ceremony - International Year of Basic Sciences for Sustainable Development \(IYBSSD\)](#)

15 Dec | Knowledge sharing | IT Amphitheatre or online | KT Group | EN
[Applied AI at CERN and Projects with Industry](#)

January

8–19 Jan | Knowledge Sharing | Main Building | Staff Association | EN/FR
[Shelter for Science photo exhibition](#)

9 Jan | Physics | Online | EuCAPT | EN
[Jocelyn Bell Burnell, "Women in Physics"](#)

11 Jan | Knowledge Sharing | IT Amphitheatre | Visit Service | EN
[Introductory course for guides](#) (see also [full list of guide training](#))

12 Jan | Knowledge Sharing | CERN Science Gateway | Public events | EN
[The Infinite Monkey Cage](#)

15 Jan | Computing | Online | HR Learning and Development | EN
[Computer security is not complete without you](#)

17 Jan | Knowledge Sharing | Main Auditorium | LHC ECSF | EN
[LHC Soft Skills Workshop: On Networking, Proactivity and Enlarging our Circles of Influence](#)

22–23 Jan | Computing | CERN Science Gateway | Voxxed Days | EN
[Voxxed days CERN](#)

26 Jan | Knowledge Sharing | IT Amphitheatre | HR Learning and Development | FR
[L&D Micro-talk 8 – Ego vs innovation : le regard des neurosciences cognitives](#)

30 Jan | Knowledge Sharing | Council Chamber | Staff Association | FR
[Projection film : CE QUI NOUS SAUVE](#)

Registration now open

9–11 Feb | Knowledge Sharing | CERN Science Gateway | Alumni Relations
[CERN Alumni 3rd collisions](#)

11–14 Feb | Accelerators | Wiener Neustadt (AT) | MedAustron, GSI, I.FAST | EN
[5th Slow Extraction Workshop](#)

This is a curated list of events relevant to the CERN community.

More events are available here: home.cern/events
Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Send a CERN e-card



Send colleagues, family and friends season's greetings using the CERN e-card service: [watch the video on YouTube](#).

2023 was another year of achievement for CERN with the excellent performance of its scientific facilities, beautiful physics results and the opening of the new centre for scientific education and outreach, the Science Gateway.

In 2024, CERN will celebrate its 70th anniversary and a rich past while advancing a vision to use its unique accelerator complex to drive 70 more years of exploration in fundamental science. The year ahead will see celebratory events and activities that connect CERN's heritage with its exciting future.

You can create your own personalised electronic cards by signing in with your CERN account [on](#)

[this site](#). You can choose from three background designs.

CERN & Society Foundation end-of-year campaign

The CERN & Society Foundation is pleased to announce its annual end-of-year campaign.

From 28 November to the end of December 2023, the CERN & Society Foundation will guide you through a digital journey in support of the 2024 non-Member State summer student programme (NMSSS).

The NMSSS provides students from non-Member States who are studying for a bachelors or masters degree with the unique opportunity to spend eight weeks at CERN. In addition to participating in experiments, students have the chance to experience CERN's multidisciplinary and multicultural environment.

During the campaign, the Foundation will tell the exciting stories of five young scientists who took part in the 2023 programme. These students were

Angelica from the Philippines, Mary from Kenya, Jenan from Palestine, Eliacim from the Dominican Republic and Nima from Iran.

Run digitally, the campaign will include six video testimonials. The first, featuring all five of the students, will give an overview of the programme and mark the campaign's launch.

All videos will be published through the CERN & Society Foundation's social media channels [Facebook](#) and [LinkedIn](#).

The CERN & Society Foundation is pleased to invite the whole CERN community to support the NMSSS programme by donating through its [website](#). We thank you in advance for supporting the next generation of scientists!

CERN & Society

Changes to the TPG public transport network from 10 December 2023

The Geneva public transport operator TPG (*Transports publics genevois*) is making a number of changes to its services with effect **from 10 December**. Certain bus routes will be renamed, new stops will be introduced and cross-border services (Pays de Gex and Haute-Savoie) will be enhanced.

Here is an overview of the changes to the bus routes serving CERN and the surrounding area:

- Creation of a new bus route [\(67\)](#) linking Meyrin Gravière with Chevry and Crozet.
- The [68 bus route](#) linking Blandonnet and the Thoiry shopping centre will be extended between the Hôpital de la Tour and Blandonnet stops, with new stops at Maillard, Veyrot and Meyrin Gare. In 2024,

this service will be gradually improved, notably with the provision of articulated buses during rush hours.

- The [F bus route](#) between Cornavin and Gex-L'Aiglette will be replaced by the [60](#) + [61 express services](#) (the 61 will run during rush hours Mondays to Fridays and will serve fewer stops).

Changes are also afoot for the Noctambus night service: most of the TPG's urban and regional routes will now operate both during the day and at night.

No changes are planned for the [18 tram line](#), which serves CERN's Meyrin site, or the [66 bus route](#), which serves the Prévessin site.

See the following link for more information:

- [Toutes les nouveautés sur le réseau TPG](#)
 - [Nouveautés lignes Rive droite / Pays de Gex](#)
 - [Pays de Gex agglo – Changements dans l'offre des transports en commun](#)
 - [Plans of the TPG network](#)
-

15 December | One-day internal workshop on applied AI at CERN

Learn about CERN's artificial intelligence (AI) driven projects with industry and exchange ideas with colleagues involved in this work

The Knowledge Transfer (KT) group would like to invite all who are interested to attend an Applied AI at CERN and Projects with Industry workshop, which will take place **all day on 15 December, in person at the IT Amphitheatre and on Zoom**.

The primary aim of the workshop is to introduce CERN's community to recent projects financed by industry (or other external organisations) using CERN's expertise and/or technology in AI and machine learning (ML). The workshop will demonstrate the richness and diversity of AI applications and explain how CERN's Knowledge Transfer group can help you to build such projects for the benefit of CERN and industry in CERN's Member and Associate Member States.

The second objective is to share ideas about future projects with the wider ML community at CERN: to

discuss common experiences, identify collaboration opportunities and learn from each other in order to optimise our internal efforts.

Finally, we are keen to open up the discussion to the CERN community and gauge the potential interest in a broader AI/ML initiative for CERN, with the aim to facilitate synergy and knowledge exchange across the Laboratory. In doing so, we aim to collectively think about how we can benefit from advancements in AI, while also taking advantage of opportunities to collaborate and partner with industry.

We are excited to build upon the vibrant communities already thriving within the Laboratory. Please join us to share your experience and ideas!

[Please register on Indico](#)

23-24 January 2024 | Overleaf Workshops: Beginners to Advanced

CERN and Overleaf, in the framework of the H2020 FCCIS project, **will offer a series of Overleaf training sessions on 23 and 24 January 2024** combined with hands-on experience at CERN's training centre. The courses are tailored to every skill level — from beginner to TeXpert! You can pick the journey that best suits you.

A **Beginner Workshop** will introduce you to Overleaf and LaTeX and show you the fundamentals of document preparation.

For those already familiar with LaTeX, the **Intermediate Workshop** will jump right into

Overleaf features and how you can use our tools to speed up your workflows.

Already completed a project on Overleaf or using LaTeX locally? Then the **Advanced Workshop** is for you! Here we will show you the Overleaf features that you may be missing out on as well as discuss best practices and more complex typesetting.

Finally, an **Extended Overview** will walk you through all of the advanced topics, remaining accessible to any level.

Pairing the Beginner/Intermediate Workshops with the Extended Overview is the perfect way to

see all the content without being overwhelmed. We hope you can join us and can't wait to show you what Overleaf can do! Participation at these trainings is free of charge, but registration is compulsory. The registration

deadline is **Monday 15 January 2024**. For more information and registration: <https://indico.cern.ch/event/1354754/>.

26 January 2024 | L&D Micro-talk: Ego vs innovation, le regard des neurosciences cognitives



Micro talk #8



HR Learning & Development vous invite à un huitième micro-talk, où nous examinerons, en nous appuyant sur les neurosciences cognitives, ce qu'est l'ego et en quoi il constitue une menace pour l'agilité de notre Organisation.

Nous nous pencherons en particulier sur l'interrelation entre l'ego et certains comportements qui empêchent l'apprentissage.



Human Resources
Learning and Development

**Ego vs innovation :
le regard des neurosciences cognitives**

Micro-Talk 60 min + 15 min Q&A

Expert : Marc Turiault, docteur en neurosciences, responsable d'équipe, formateur et conférencier.

26 janvier 2024 - 14h - 15 h 15

En français et en présentiel au CERN dans l'Amphithéâtre IT 31/3-004
Zoom/webcast avec enregistrement (sauf la séance de questions et les échanges personnels)
<https://indico.cern.ch/event/1335703/>

CERN's Office of Data Privacy releases the third video in its "Privacy in a nutshell" series

Have you ever wondered about your role as a data subject in the world of data privacy? It's not just about regulations; it's also about your rights and responsibilities.

In our ongoing effort to make data privacy more accessible, the Office of Data Privacy (ODP) at CERN, in partnership with the Data Privacy Coordination Committee (DPCC), presents the

next instalment in our “Privacy in a nutshell” video series.

[This third episode](#) explores the concept of data subjects, shedding light on what being a data subject means whilst clarifying your rights and responsibilities. Discover how CERN protects your personal data and learn what being a data subject entails.

For more resources and related information, please visit the [ODP website](#).

Stay tuned for more informative videos as we continue our mission to make data privacy understandable and relevant to all.

Office of Data Privacy

Dismantling of work around R1 extension to take place over end-of-year closure

Since September, the Restaurant 1 extension has been undergoing consolidation work. This took place to repair its roof overhangs and to secure the glass doors that frequently broke, which compromised public safety and often required costly interventions. In this framework, the SCE department has undertaken extensive work, which highlighted other unforeseen issues and further consolidation needs. This extended the work beyond the two-initially-planned months. The work has now been completed. However, the removal of the scaffolding and protective covering, which acted as an umbrella over the construction, will require the use of a crane. To limit the impact of the crane installation on the

parking des Cèdres, ensure totally safe operations and reduce inconvenience for nearby occupants and users, the SCE department has chosen to dismantle the whole installation during the annual end-of-year closure.

The work installations will remain until the end of the year. The restaurant extension will remain fully accessible to users, as it has been throughout this period.

The SCE department thanks you for your understanding and apologises for any inconvenience.

SCE department

Become an ambassador for Women and Girls in Science and Technology!

For the eighth year running, CERN, the UNIGE Faculty of Science, EPFL and the Laboratoire d'Annecy de Physique des Particules (LAPP) are joining forces to celebrate the International Day of Women and Girls in Science. During the week of 5 to 9 February 2024, female scientists and engineers will be visiting local schools to inspire younger generations to explore the world of science.

As a female science ambassador, you will talk about your career path, share your projects and

professional experiences and maybe even give a short demonstration. Our aim is to change the way schoolchildren view scientific, technical and technological professions, making them accessible to both girls and boys. And who knows, these presentations might even inspire some future careers!

Every year, the Women and Girls in Science and Technology week proves to be a great success. In 2023, over 240 presentations were given by 100 or so female science ambassadors to 5220

schoolchildren! **This is why we're always looking for more female scientists willing to give up a bit of their time to give talks in schools.** So come and join the adventure by [signing up](#): **the deadline is 10 December 2023 (11.59 p.m.).**

Conditions of participation:

- Registration is open to all women working in a profession connected with science, technology, engineering or maths (STEM), as well as computer science, communication or education.
- Priority will be given to presenters from CERN, UNIGE, EPFL and LAPP. If you are not from one of these institutions but would like to take part, please [contact us](#).
- Volunteers will deliver one-hour presentations for a maximum of 35 pupils aged 7 to 15.

- The majority (95%) of the presentations will be given in French, but presentations in English are also possible.
- Volunteers are required to attend a briefing session.

Sign up and find out more: <http://cern.ch/fds-interne>

Deadline for signing up: 10 December 2023 (11.59 p.m.)

For CERN volunteers only: we are also looking for a female scientist who is available on Saturday, 24 February at 10.30 a.m. to talk (in French) to children aged 7 and over about her job and about particle physics in general. This event is being organised by the Bourg-en-Bresse multimedia library. Transport expenses will be covered. For more information and to volunteer, contact education.locale@cern.ch.

Thank you for volunteering!

Obituaries

Philippe Bernard (1935 – 2023)



Born in 1935, Philippe Bernard completed his studies at the prestigious *Ecole supérieure d'électricité* in 1956. He began working at CERN in 1962 as engineer-in-charge of the Proton Synchrotron. He went on to design and develop radiofrequency (RF)

separators, making substantial contributions to the improvement of these devices that provide well-selected secondary beams. This was particularly important in the early 1970s for experiments with the CERN 2m hydrogen chamber, the Saclay-built Mirabelle chamber at IHEP in Serpukhov, USSR, and the Big European Bubble Chamber (BEBC) at CERN.

Already with these separators, he realised the potential of superconductivity for RF structures,

and so was entrusted by John Adams in 1978, together with Herbert Lengeler, to develop RF cavities for CERN accelerators. A vigorous programme, with international participation, led to the development of five-cell cavities, first made of pure niobium and, later, of niobium sputtered on the more stable copper-substrate to produce robust cavities. This allowed accelerating fields of up to 7 MV/m to be reached.

After tests of prototypes at Petra (DESY) and the Super Proton Synchrotron, 320 such cavities were produced for the Large Electron-Positron collider (LEP) using niobium-film technology. In the framework of the upgrade programme, which started in 1987, these cavities were gradually added to the complement of normal-conducting cavities, which were partially replaced. This enabled an increase in the electron and positron beam energy from 46 GeV in 1989 to 104 GeV by 2000. In addition to this successful development, in the late 1990s Philippe took a strong interest in the design and development of a system of

coupled superconducting cavities as a sensitive detector of gravitational waves.

Philippe was also involved in numerous CERN-wide activities, including chairing the purchasing policy monitoring board and serving as president of the CERN Health Insurance Scheme (CHIS). He also served as president and vice-president of the CERN Pensioners Association during a critical period.

His open mind, his wide-ranging views and his solid technical knowledge made him a recognised

leader. His critical and thoughtful attitude made him a respected discussion partner for the CERN Management. His commitment to the CERN Health Insurance Scheme and to long-term improvements in the social conditions of CERN and ESO staff was widely appreciated and acknowledged. We remember him as a generous, witty and vivacious friend.

His friends and colleagues

Ombud's corner

Impartiality or, better, multi-partiality?

Confidentiality, informality, independence, neutrality and impartiality are the working principles of the Ombud and I have shared with you in a number of articles my experience in putting these principles into practice.

With significant experience today in the role of CERN Ombud, I would like to come back to neutrality and impartiality and propose a somewhat different view of how these principles are embedded in my Ombud's practice.

The Ombud is a designated **neutral**, which means that the Ombud does not engage in any situation that would create a conflict of interest. This is largely made possible and supported by the Ombud's independence.

Impartiality, on the other hand, refers to an absence of bias. It relates directly to not favouring one side's interests over the other side's interests. The requirement to remain impartial does not force ombuds to refrain from having personal opinions or from sympathising more with one party over another. Instead, this principle prompts ombuds to avoid acting on those preferences and requires them to separate their personal opinions from the performance of their duties.

When I took on the role of Ombud, I found the principle of impartiality to be the most challenging one. Not only are we, as human beings, wired to take sides, but the very idea of not choosing sides

often raises suspicion. In addition, it seemed to me that impartiality asked me to disengage from the parties and to maintain a solid wall between myself and them.

Recently, I discovered that being engaged and simultaneously maintaining an equidistant stance between the parties has a name: multi-partiality. And I feel very comfortable with this concept. Rather than not favouring anyone, being multi-partial means that I favour everyone equally and I engage – with my intellect, emotions, energy, knowledge and skills – **equally with everyone**.

In the vast majority of the conflicts that my visitors share with me, I hear only their perception of the situation, as I rarely meet the other party.

However, it may also happen that both parties in a conflict come, independently, to discuss the situation with me, without either of them being aware that I have access to the other side of the story.

These situations are prime opportunities for the Ombud to exercise multi-partiality to the full. It is amazing to see how the same situation may be described differently by two parties. The two situations exposed may only intersect – at best – with a few facts, as not all factual information is retained by one party. The feelings, the emotions that one party shares are completely misunderstood by the other party. Words are

interpreted differently, intentions are misunderstood. And the impact that the situation has on the other party is totally ignored.

When parties in a conflict have agreed to engage in a mediation with the Ombud, they often realise how differently they have experienced a situation and how their own anger, bitterness and perplexity have arisen from this misunderstanding.

My personal biases come from my experience, history and character. Impartiality, defined as the absence of bias, may be very challenging to exercise. For example, I may be instantly triggered by sexist remarks or blunt bad faith.

Multi-partiality, however, allows me to engage fully with each party while maintaining an equidistant stance between them.

At this time of the year, I would like to thank you, readers, for your interest in the Ombud's corner. I wish you all a very relaxing end of the year break with your dear ones and look forward to exchanging with you in 2024 with renewed energy!

I would like also to mention that, **with my upcoming retirement in spring next year, [a vacancy notice](#)** for the Ombud's position has been published. If you are interested and would like to discuss the role with me, you are most welcome to contact me!

Laure Esteveny